

Chapter 4

Work and Kinetic Energy

Introduction

- A baseball pitcher does work with his throwing arm to give the ball a property called **kinetic energy**.
- In this chapter, the introduction of the concepts of *work*, *energy*, and the *conservation of energy* will allow us to deal with problems in which Newton's laws alone aren't enough.



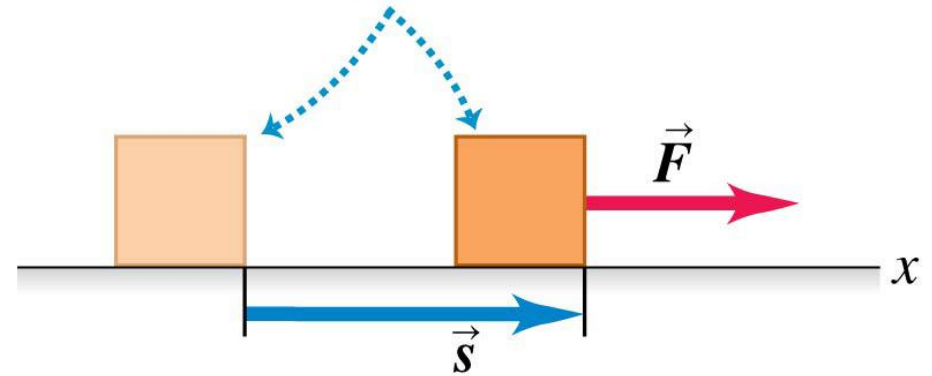
Work

- A force on a body does *work* if the body undergoes a displacement.



These people are doing work as they push on the car because they exert a force on the car as it moves.

If a body moves through a displacement $\vec{s}$ while a constant force $\vec{F}$ acts on it in the same direction ...



... the work done by the force on the body is $W = Fs$.

Units of work

- The SI unit of work is the **joule** (named in honor of the 19th-century English physicist James Prescott Joule).
- Since $W = Fs$, the unit of work is the unit of force multiplied by the unit of distance.
- In SI units:

$$1 \text{ joule} = (1 \text{ newton}) (1 \text{ meter}) \text{ or } 1 \text{ J} = 1 \text{ N} \cdot \text{m} = 1 \text{ kg m}^2/\text{s}^2$$

- If you lift an object with a weight of 1 N a distance of 1 m at a constant speed, you do 1 J of work on it.



Work done by a constant force

- The work done by a constant force acting at an angle ϕ to the displacement is:

Work done on a particle by constant force $\vec{F}$ during straight-line displacement $\vec{s}$

$$W = Fs \cos \phi$$

Magnitude of $\vec{F}$
Angle between $\vec{F}$ and $\vec{s}$
Magnitude of $\vec{s}$

- This can be written more compactly as:

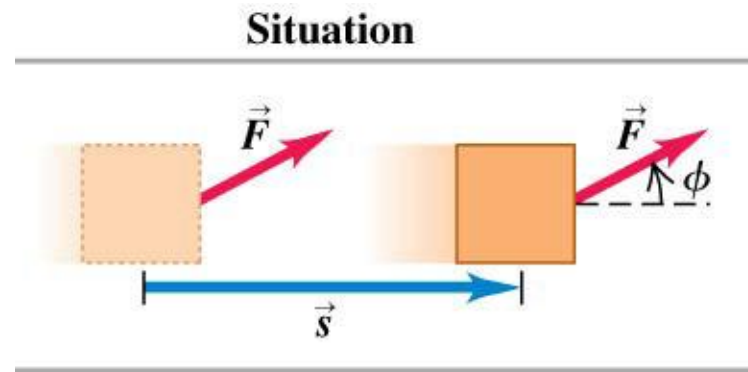
Work done on a particle by constant force $\vec{F}$ during straight-line displacement $\vec{s}$

$$W = \vec{F} \cdot \vec{s}$$

Scalar product (dot product) of vectors $\vec{F}$ and $\vec{s}$

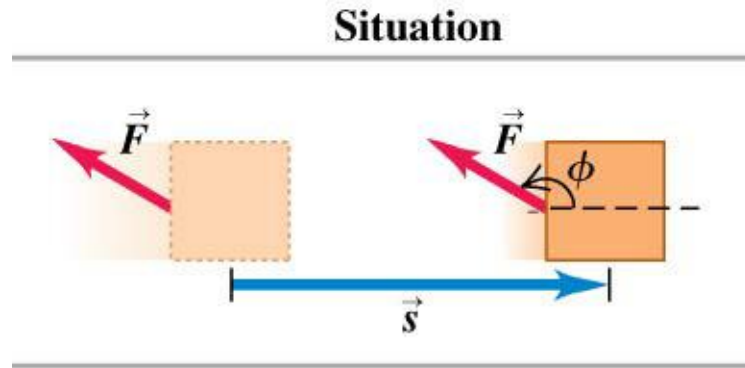
Positive work

- When the force has a component in the direction of the displacement, work is *positive*.



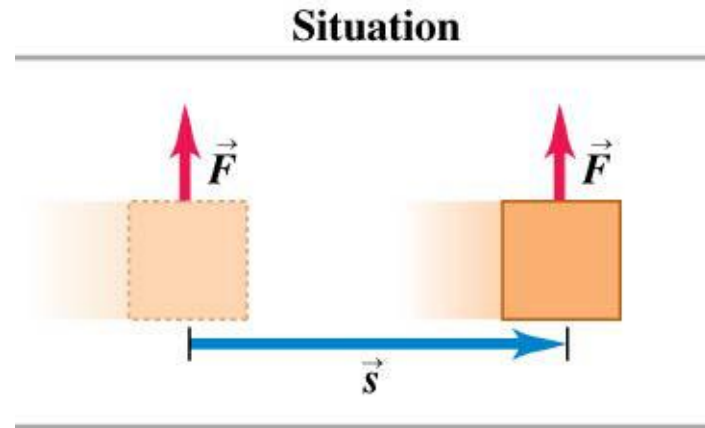
Negative work

- When the force has a component opposite to the direction of the displacement, work is *negative*.



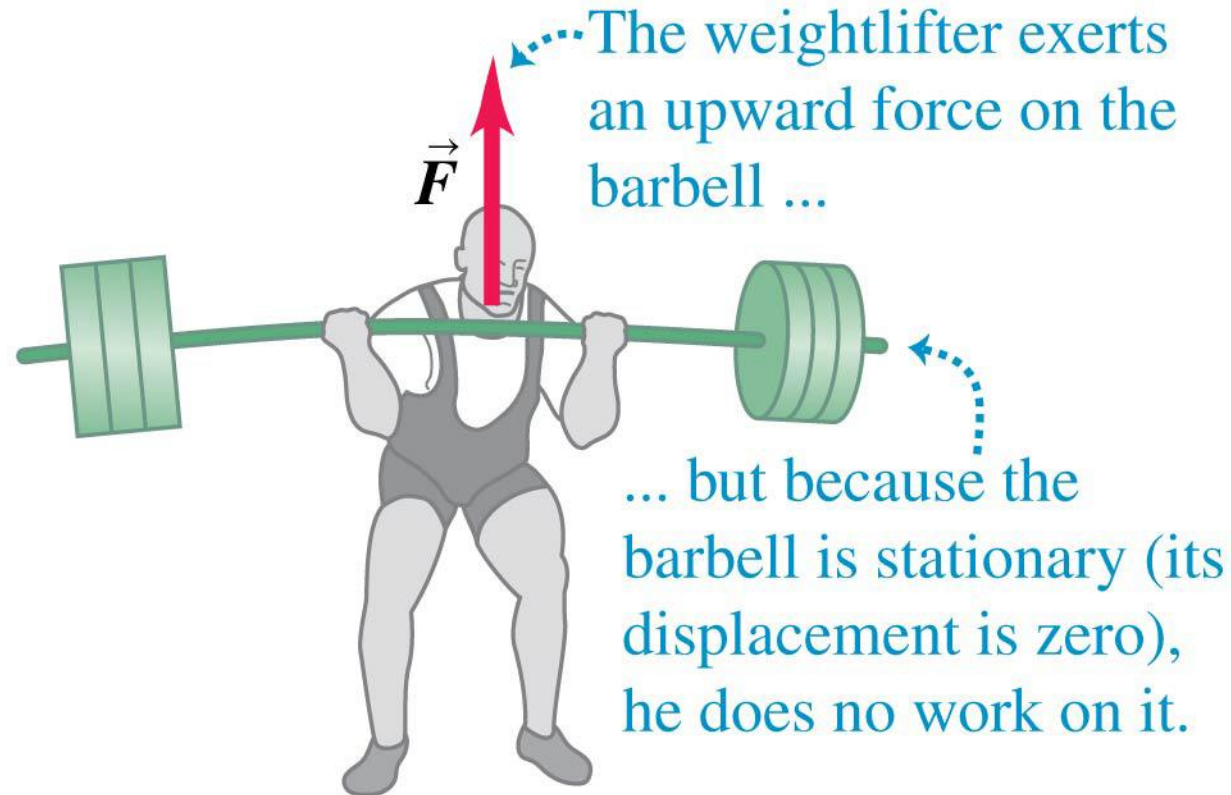
Zero work

- When the force is perpendicular to the direction of the displacement, the force does *no* work on the object.



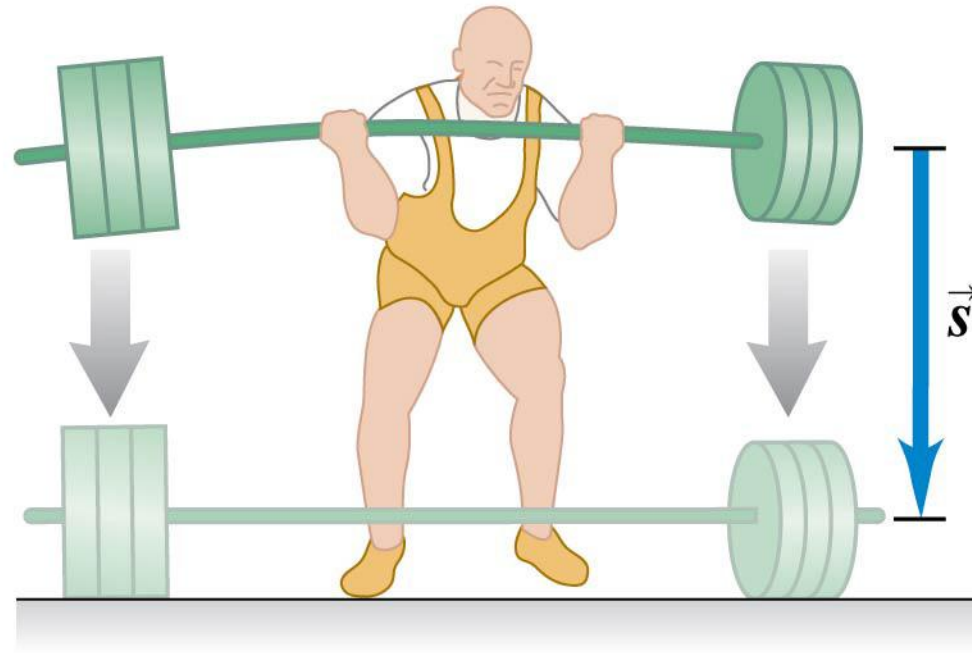
Zero work

- A weightlifter does no work on a barbell as long as he holds it stationary.



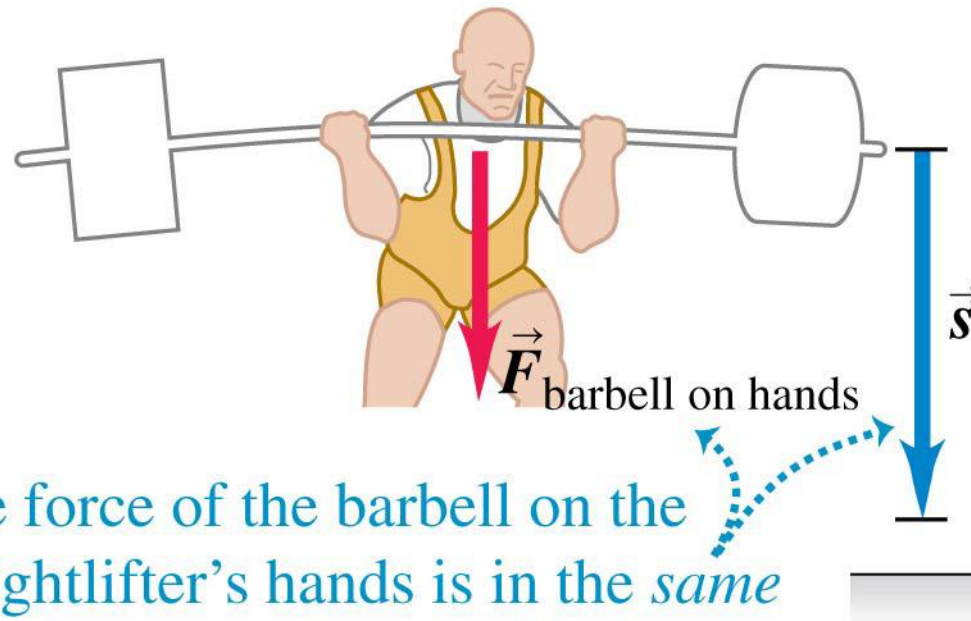
Lowering the barbell to the floor: Slide 1

A weightlifter lowers a barbell to the floor.



Lowering the barbell to the floor: Slide 2

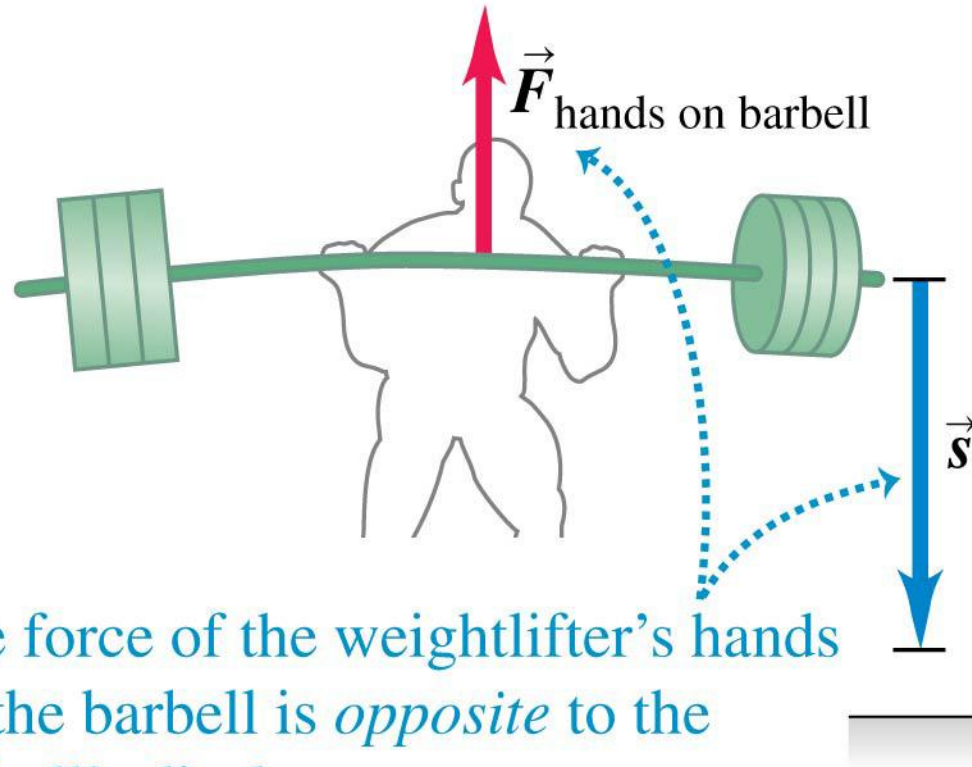
The barbell does *positive* work on the weightlifter's hands.



The force of the barbell on the weightlifter's hands is in the *same* direction as the hands' displacement.

Lowering the barbell to the floor: Slide 3

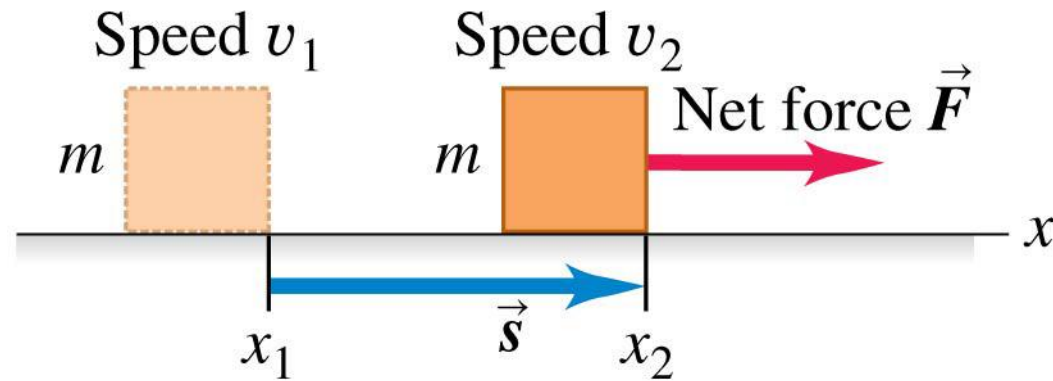
The weightlifter's hands do *negative* work on the barbell.



The force of the weightlifter's hands on the barbell is *opposite* to the barbell's displacement.

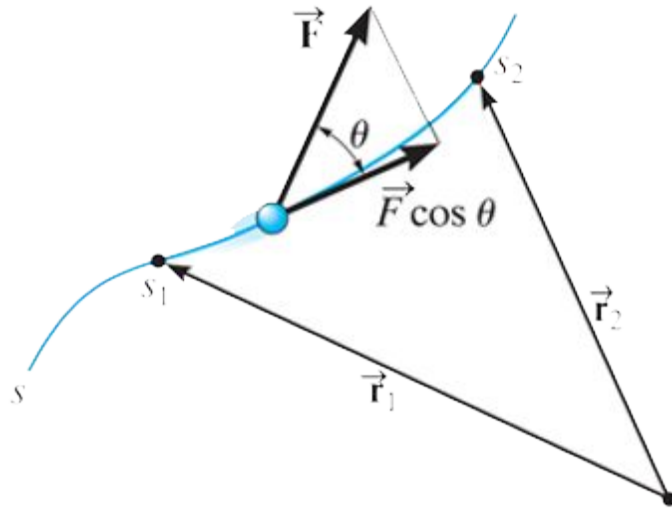
Total work

- The work done by the net force on a particle as it moves is called the **total work** W_{tot} .
- The particle speeds up if $W_{\text{tot}} > 0$, slows down if $W_{\text{tot}} < 0$, and maintains the same speed if $W_{\text{tot}} = 0$.



Work of a force: general formalism

A force does **work** on a particle when the particle undergoes a **displacement along the line of action of the force**.



Work is defined as the **product of force and displacement components** acting in the **same direction**. So, if the angle between the force and displacement vector is θ , the increment of work dW done by the force is

$$dW = F ds \cos \theta = \vec{F} \cdot d\vec{r}$$

By using the definition of the **dot product** and integrating, the total work can be written as

$$W_{1-2} = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r}$$

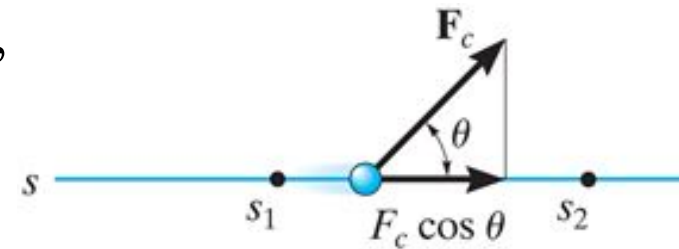
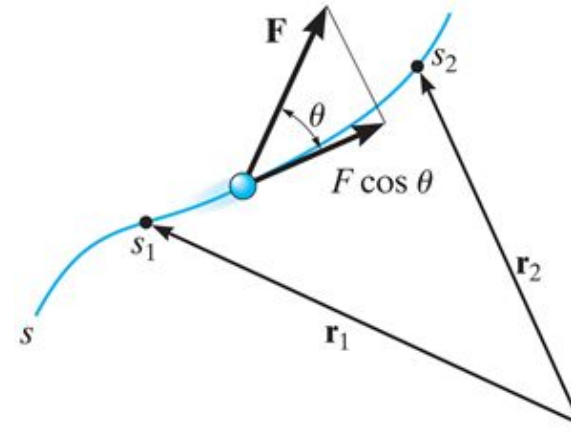
Work of a force: general formalism

If $\mathbf{F}$ is a function of position (a common case) this becomes

$$W_{1-2} = \int_{s_1}^{s_2} F \cos \theta \, ds$$

If both F and θ are constant ($F = F_c$), this equation further simplifies to

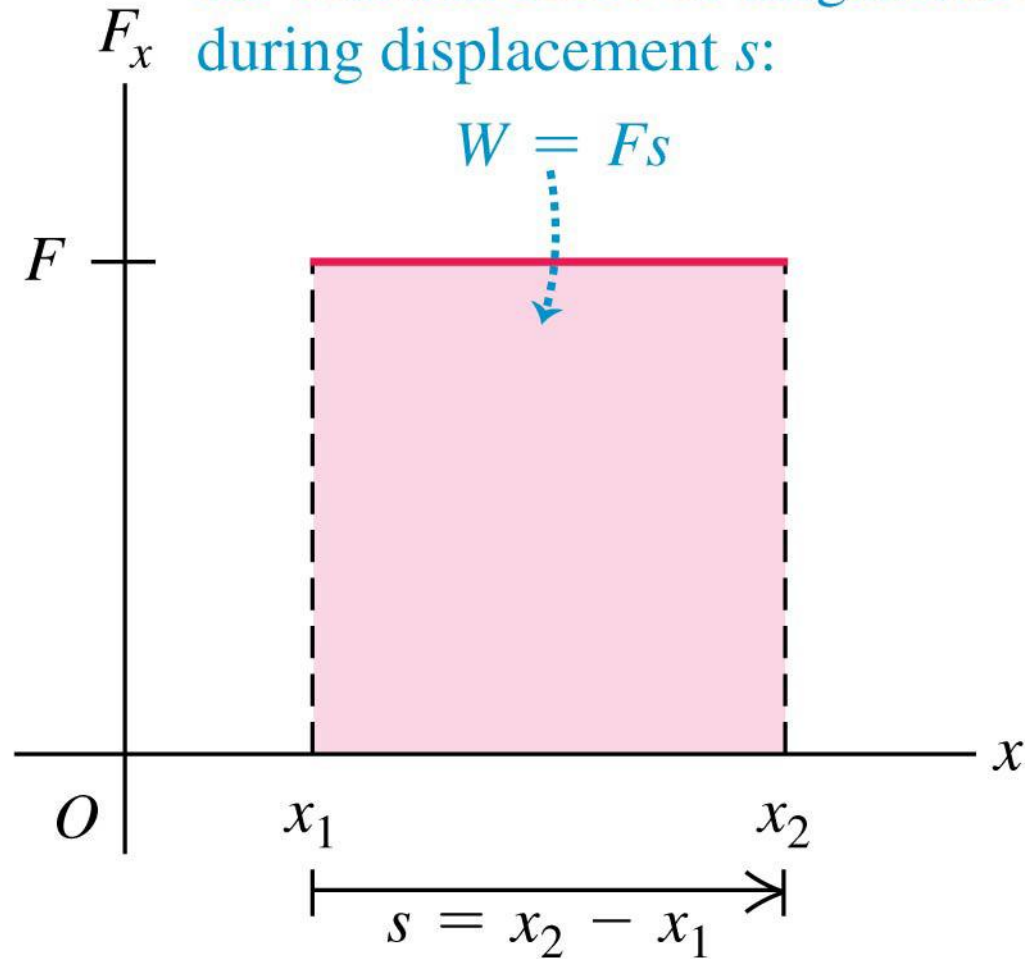
$$W_{1-2} = F_c \cos \theta (s_2 - s_1)$$



Work is **positive** if the force and the movement are in the **same direction**. If they are **opposing**, then the work is **negative**. If the force and the displacement directions are **perpendicular**, the work is **zero**.

Work done by a constant force

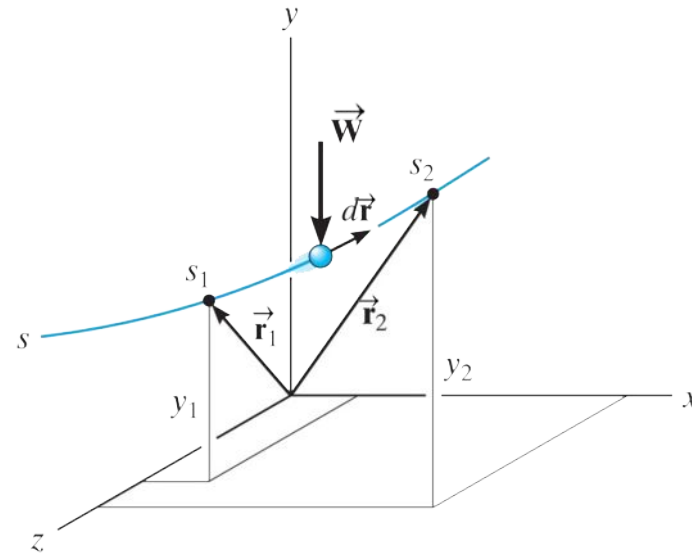
The rectangular area under the graph represents the work done by the constant force of magnitude F during displacement s :



Work of a weight

The work done by the gravitational force acting on a particle (or **weight of an object**) can be calculated by using

$$\begin{aligned} W_{1-2} &= \int_{\vec{r}_1}^{\vec{r}_2} m\vec{g} \cdot d\vec{r} = \int_{\vec{r}_1}^{\vec{r}_2} -mg\hat{y} \cdot d\vec{r} \\ &= \int_{\vec{r}_1}^{\vec{r}_2} -mgd(\hat{y} \cdot \vec{r}) = \int_{y_1}^{y_2} -mgdy \\ &= -mg(y_2 - y_1) = -mg\Delta y \end{aligned}$$



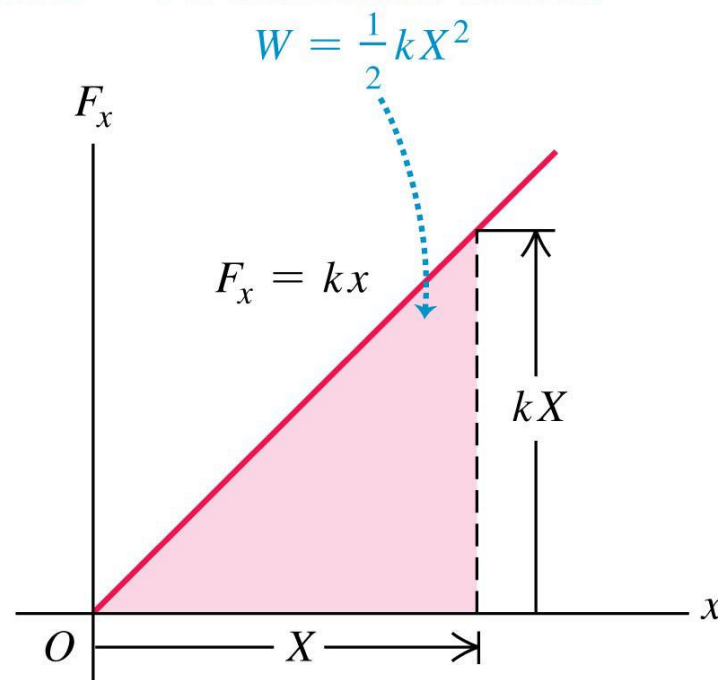
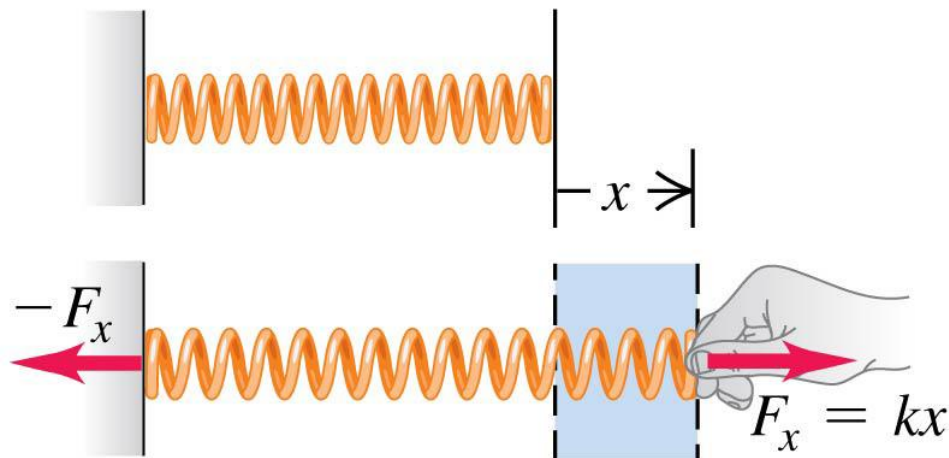
The work of a weight is the product of the magnitude of the particle's weight and its vertical displacement. If Δy is **upward**, the work is **negative** since the weight force always acts downward.

Stretching a spring

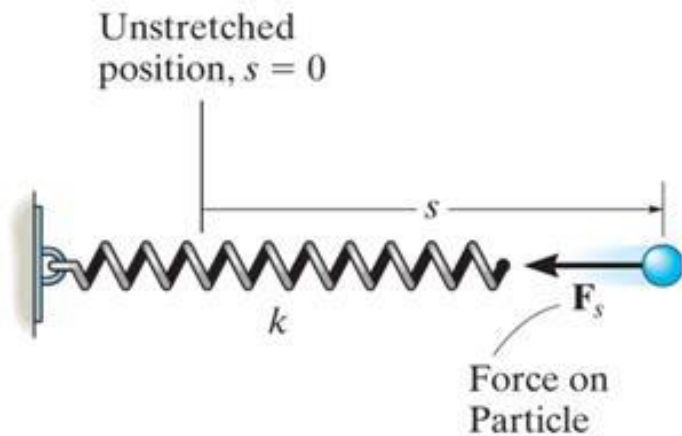
- The force required to stretch a spring a distance x is proportional to x : $F_x = kx$.
- The area under the graph represents the work done on the spring to stretch it a distance X :

$$W = \int_0^x kx' dx' = \frac{1}{2} kx'^2 \Big|_0^x = \frac{1}{2} kx^2$$

The area under the graph represents the work done on the spring as the spring is stretched from $x = 0$ to a maximum value X :



Work of a spring force



When stretched, a **linear elastic spring** develops a force of magnitude $F_s = ks$, where k is the **spring stiffness** and s is the **displacement from the unstretched position**.

If a particle is attached to the spring, the force F_s exerted **on the particle is opposite** to that exerted on the spring. Thus, the work done on the particle by the spring force will be **negative** or

$$W_{1-2} = - \left[\frac{1}{2} k (s_2)^2 - \frac{1}{2} k (s_1)^2 \right] .$$

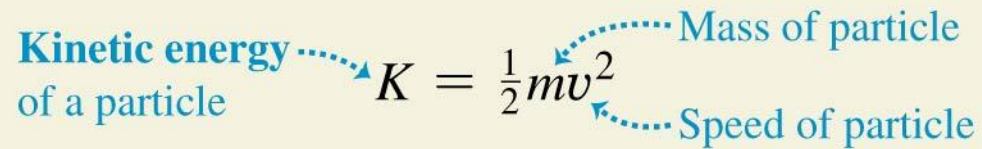
Spring forces

It is important to note the following about spring forces.

1. The equations above are for **linear** springs only! Recall that a linear spring develops a force according to $F = ks$ (essentially the equation of a line).
2. The work of a spring is **not** just spring force times distance at some point, i.e., $(ks_i)(s_i)$. **Beware**, this is a trap that students often fall into!
3. Always **double-check** the sign of the spring work after calculating it. It is positive work if the force on the object by the spring and the movement are in the same direction.

Kinetic energy

- The energy of motion of a particle is called **kinetic energy**:



The diagram shows the formula $K = \frac{1}{2}mv^2$ on a light yellow background. A blue dotted arrow points from the text "Kinetic energy of a particle" to the letter K . Another blue dotted arrow points from the text "Mass of particle" to the letter m . A third blue dotted arrow points from the text "Speed of particle" to the v^2 term.

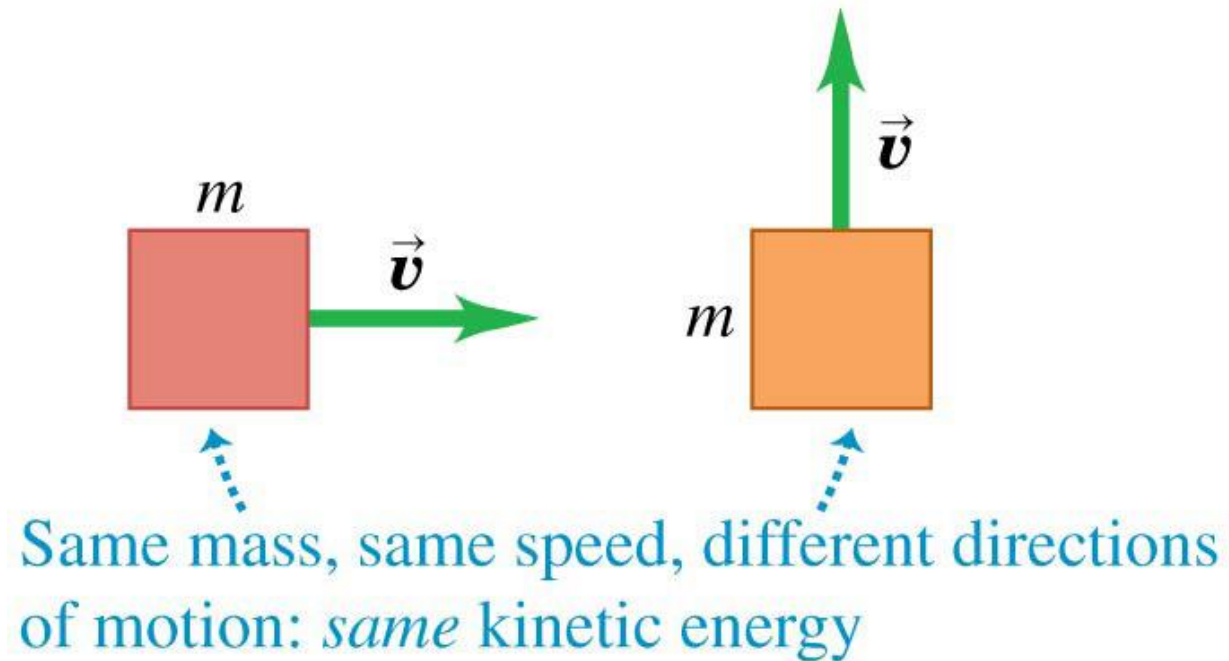
$$K = \frac{1}{2}mv^2$$

- Like work, the kinetic energy of a particle is a scalar quantity; it depends on only the particle's mass and speed, not its direction of motion.
- Kinetic energy can never be negative, and it is zero only when the particle is at rest.
- The SI unit of kinetic energy is the joule.

$$1 \text{ J} = 1 \text{ N} \cdot \text{m} = 1 \text{ kg m}^2/\text{s}^2$$

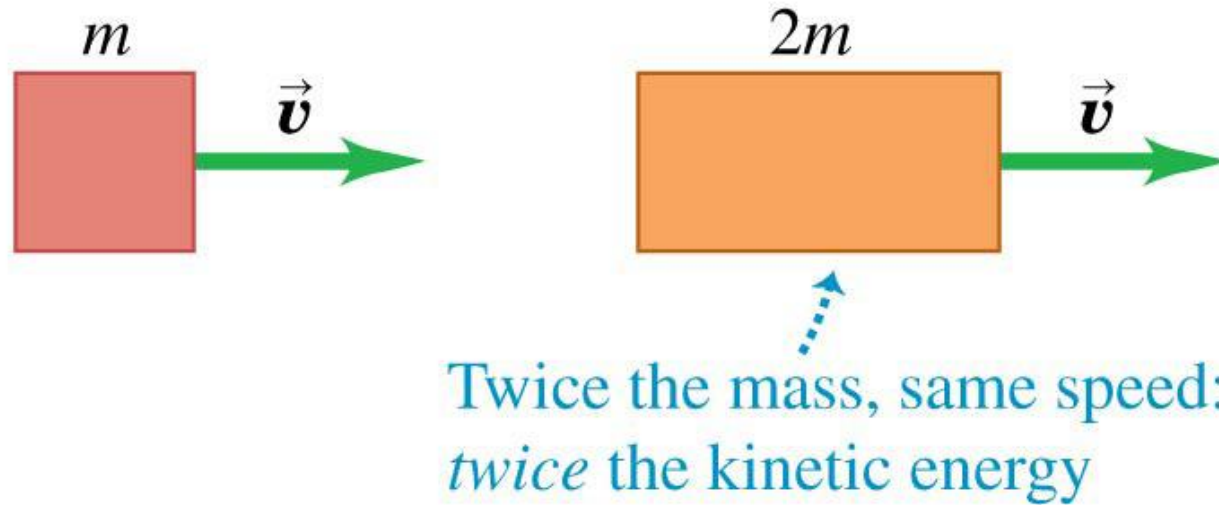
Kinetic energy

- Kinetic energy does not depend on the direction of motion.



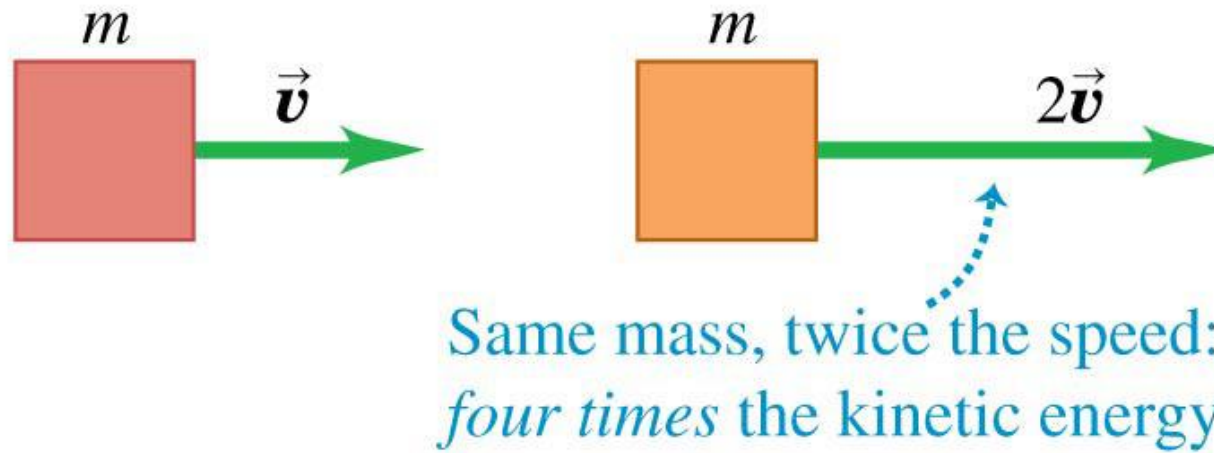
Kinetic energy

- Kinetic energy increases linearly with the mass of the object.



Kinetic energy

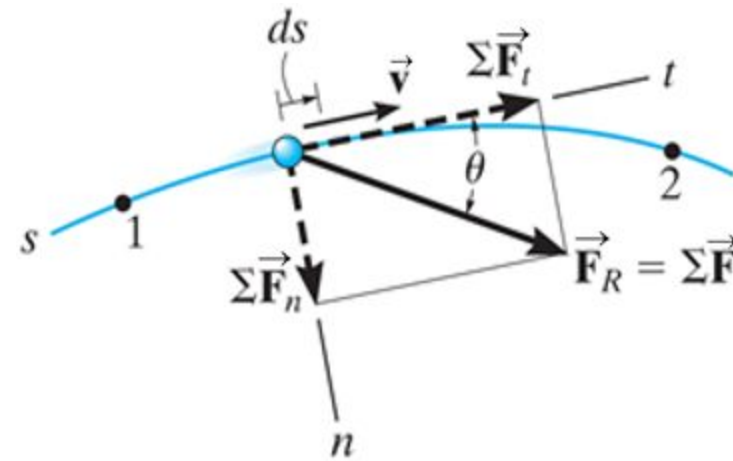
- Kinetic energy increases with the *square* of the speed of the object.



Work and energy

Another equation for working kinetics problems involving particles can be derived by **integrating** the **equation of motion** ($F = ma$) with respect to **displacement**.

By substituting $a_t = v(dv/ds)$ into $F_t = ma_t$, the result is integrated to yield an equation known as the **principle of work and energy**.



This principle is useful for solving problems that involve **force**, **velocity**, and **displacement**. It can also be used to explore the concept of **power**.

Principle of work and energy

By integrating the equation of motion, $\sum F_t = ma_t = mv(dv/ds)$, the **principle of work and energy** can be written as

$$\int_{s_1}^{s_2} F_t ds = \int_{v_1}^{v_2} mv dv$$

$$\sum W_{1-2} = 1/2 m (v_2)^2 - 1/2 m (v_1)^2 \quad \text{or} \quad K_1 + \sum W_{1-2} = K_2$$

$\sum W_{1-2}$ is the **work done by all the forces** acting on the particle as it moves from point 1 to point 2. Work can be either a **positive or negative** scalar.

K_1 and K_2 are the **kinetic energies** of the particle at the initial and final position, respectively. Thus, $K_1 = 1/2 m (v_1)^2$ and $K_2 = 1/2 m (v_2)^2$. The kinetic energy is always a **positive scalar** (velocity is squared!).

So, the particle's initial kinetic energy plus the work done by all the forces acting on the particle as it moves from its initial to final position is equal to the particle's final kinetic energy.

Principle of work and energy

Note that the principle of work and energy ($K_1 + \sum W_{1-2} = K_2$) is **not a vector equation**! Each term results in a scalar value.

Both kinetic energy and work have the same units, that of energy! In the SI system, the unit for energy is called a **joule** (J), where $1 \text{ J} = 1 \text{ N}\cdot\text{m}$. In the FPS system, units are $\text{ft}\cdot\text{lb}$.

The principle of work and energy **cannot** be used, in general, to determine forces directed **normal** to the path, since these forces do no work.

The principle of work and energy can also be applied to a **system of particles** by summing the kinetic energies of all particles in the system and the work due to all forces acting on the system.

The work-energy theorem

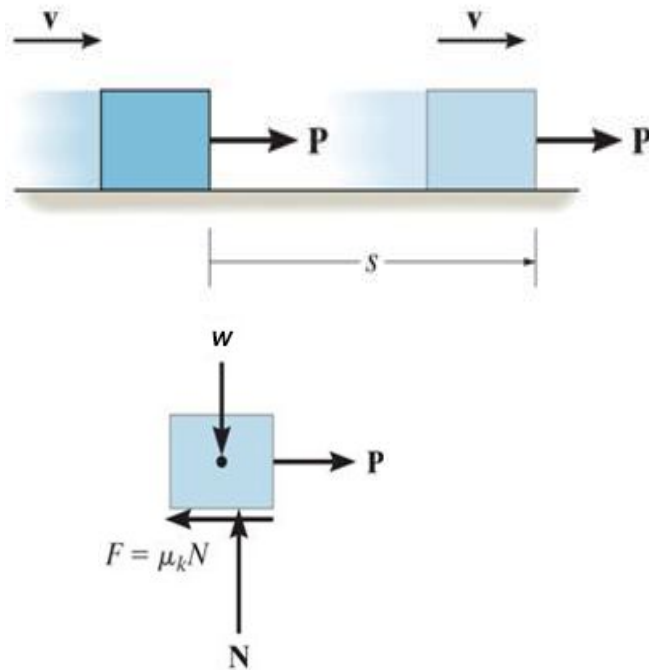
- The **work-energy theorem**: The work done by the net force on a particle equals the change in the particle's kinetic energy.

Work-energy theorem: Work done by the net force on a particle equals the change in the particle's kinetic energy.

Total work done
on particle = $\cdots \rightarrow W_{\text{tot}} = K_2 - K_1 = \Delta K \leftarrow \cdots$ Change in
work done by
net force Final kinetic energy Initial kinetic energy

Work of friction caused by sliding

The case of a body sliding over a **rough surface** merits special consideration.



Consider a block which is moving over a rough surface. If the applied force P just balances the resultant **frictional force** $\mu_k N$, a constant velocity v would be maintained.

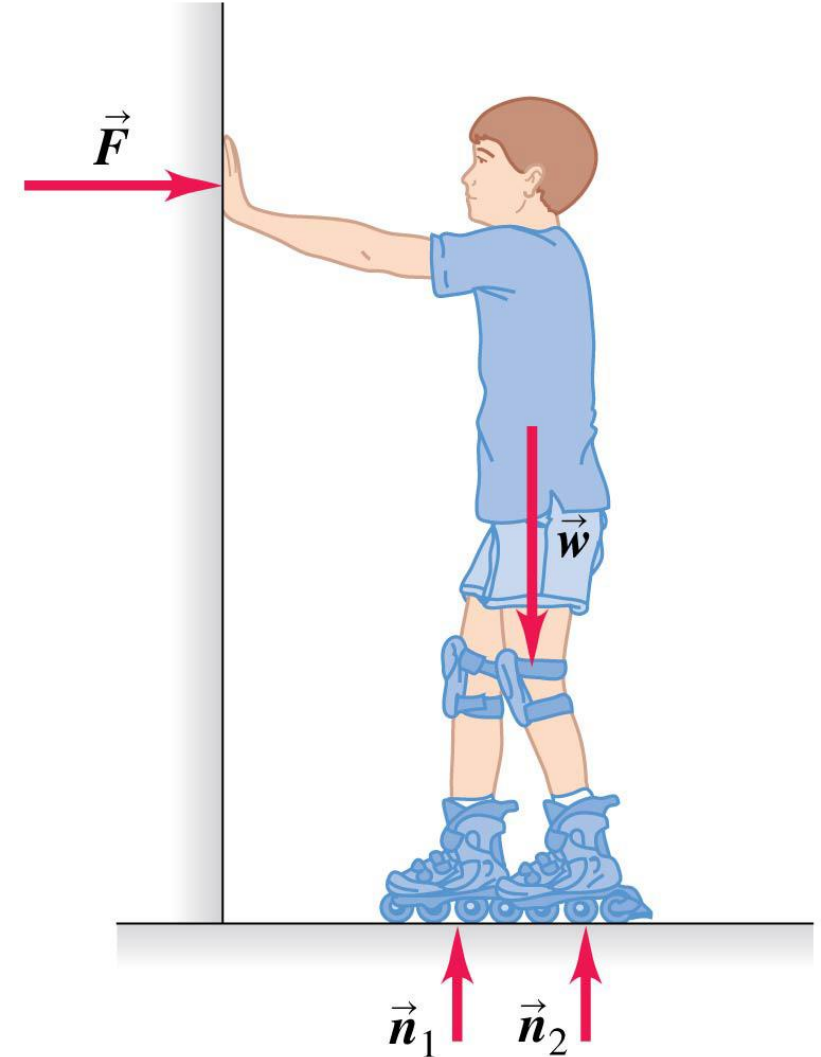
The principle of work and energy would be applied as

$$0.5m (v)^2 + P s - (\mu_k N) s = 0.5m (v)^2$$

This equation is satisfied if $P = \mu_k N$. However, we know from experience that friction generates **heat**, a form of energy that does not seem to be accounted for in this equation. It can be shown that the work term $(\mu_k N)s$ represents **both** the **external work** of the friction force and the **internal work** that is converted into heat.

Work and kinetic energy in composite systems

- The work done by the external forces acting on the skater is *zero*.
- But the skater's kinetic energy changes nonetheless!
- The explanation is that it's not adequate to represent the boy as a single point mass.



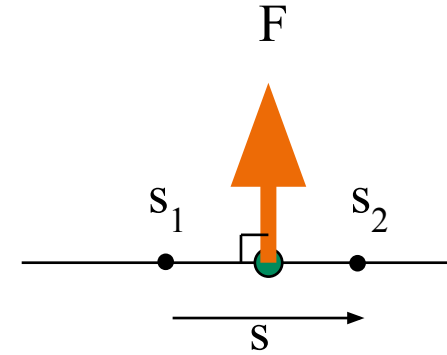
1. What is the work done by the force F ?

A) $F s$

B) $-F s$

C) Zero

D) None of the above.



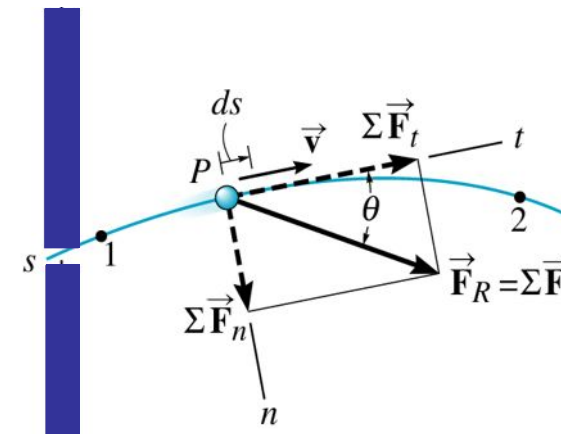
2. If a particle is moved from 1 to 2, the work done on the particle by the force, F_R will be

A)

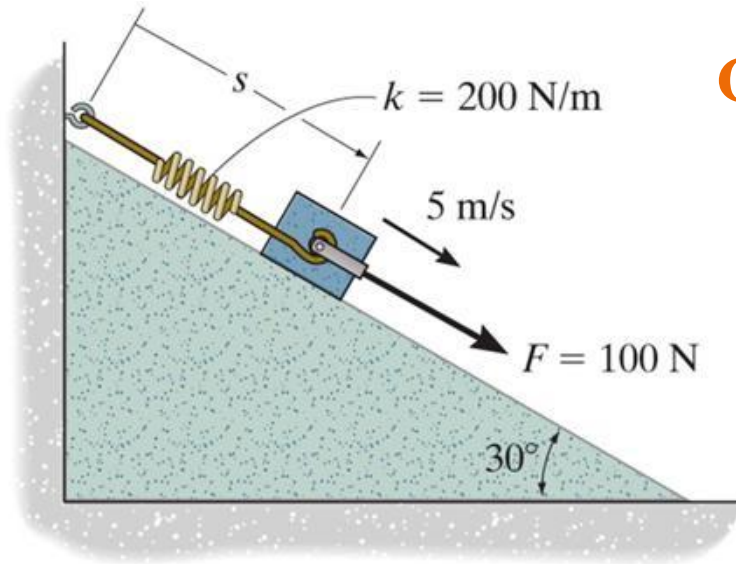
B)

C)

D)



Example



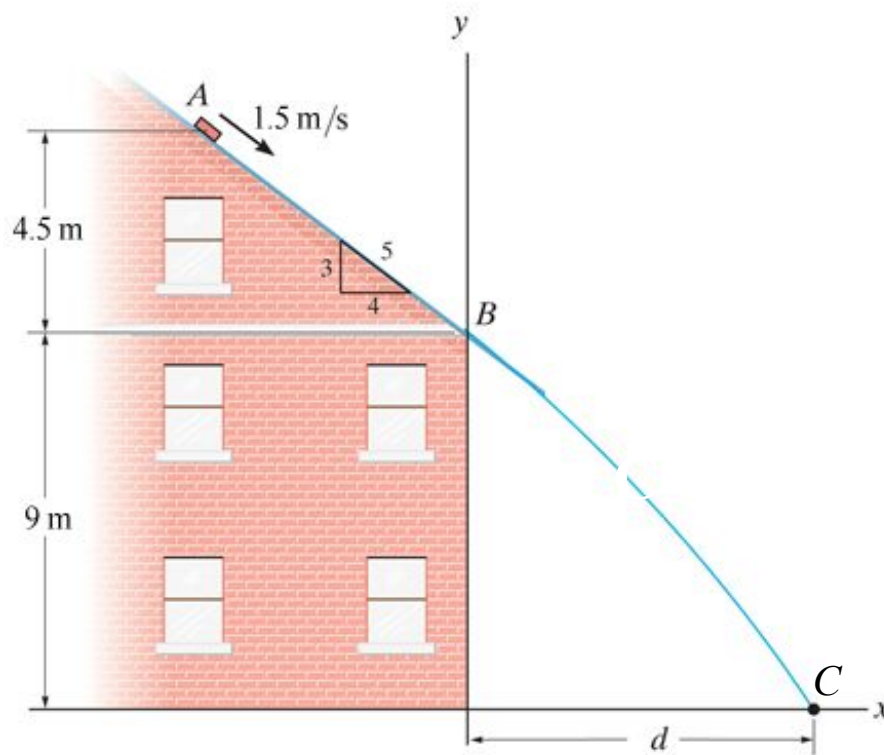
Given: When $s = 0.6 \text{ m}$, the spring is not stretched or compressed, and the 10-kg block, which is subjected to a force of 100 N , has a speed of 5 m/s down the smooth plane.

Find: The distance s when the block stops.

Quiz

1. A spring with an unstretched length of 50 mm expands from a length of 20 mm to a length of 40 mm. The work done on the spring is _____ N·mm.
- A) $-[0.5 k(40 \text{ mm})^2 - 0.5 k(20 \text{ mm})^2]$
 - B) $0.5 k (20 \text{ mm})^2$
 - C) $-[0.5 k(30 \text{ mm})^2 - 0.5 k(10 \text{ mm})^2]$
 - D) $0.5 k(30 \text{ mm})^2 - 0.5 k(0 \text{ mm})^2$
2. If a spring force is $F = 5 s^3 \text{ N/m}$ and the spring is compressed by $s = 0.5 \text{ m}$, the work done on a particle attached to the spring will be
- A) $0.625 \text{ N} \cdot \text{m}$
 - B) $-0.625 \text{ N} \cdot \text{m}$
 - C) $0.0781 \text{ N} \cdot \text{m}$
 - D) $-0.0781 \text{ N} \cdot \text{m}$

Example



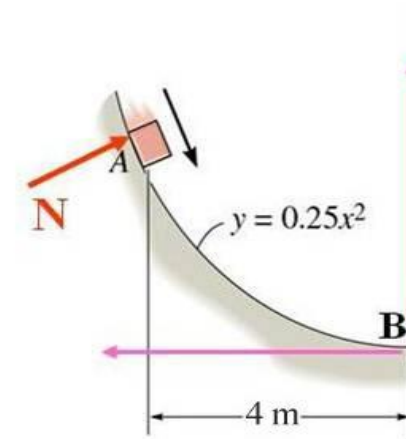
Given: The 1-kg brick slides down a smooth roof, with $v_A = 1.5 \text{ m/s}$.

Find: The speed at B, the distance d from the wall to where the brick strikes the ground (point C), and its speed at C.

Quiz

What is the work done by the normal force N if a 2-kg box is moved from A to B?

- A) $78.5 \text{ N} \cdot \text{m}$ B) $0 \text{ N} \cdot \text{m}$
C) $78.5 \text{ N} \cdot \text{m}$ D) $157 \text{ N} \cdot \text{m}$



Power

- **Power** is the *rate* at which work is done.
- **Average power** is:

$$\text{Average power during time interval } \Delta t \rightarrow P_{\text{av}} = \frac{\Delta W}{\Delta t}$$

Work done during time interval
Duration of time interval

- **Instantaneous power** is:

$$\text{Instantaneous power} \rightarrow P = \lim_{\Delta t \rightarrow 0} \frac{\Delta W}{\Delta t} = \frac{dW}{dt}$$

Time rate of doing work
Average power over infinitesimally short time interval

Horse Power

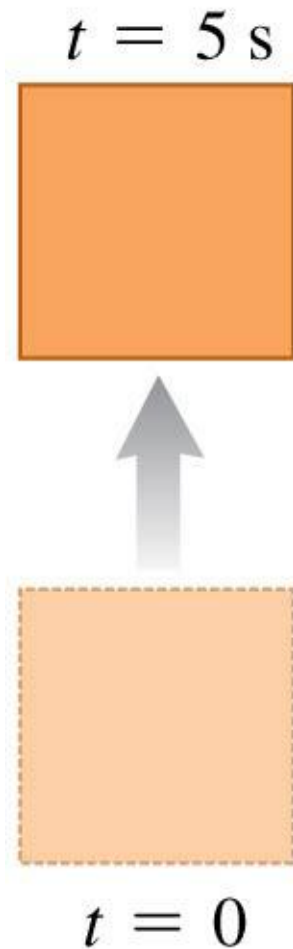
- The SI unit of power is the **watt**,

$$1 \text{ W} = 1 \text{ J/s} = 1 (\text{N} \cdot \text{m})/\text{s} = 1 \text{ kg} \cdot \text{m}^2 / \text{s}^3$$

- But another familiar unit is the *horsepower* (1 hp = 746 W).



Power: Lifting a box slowly



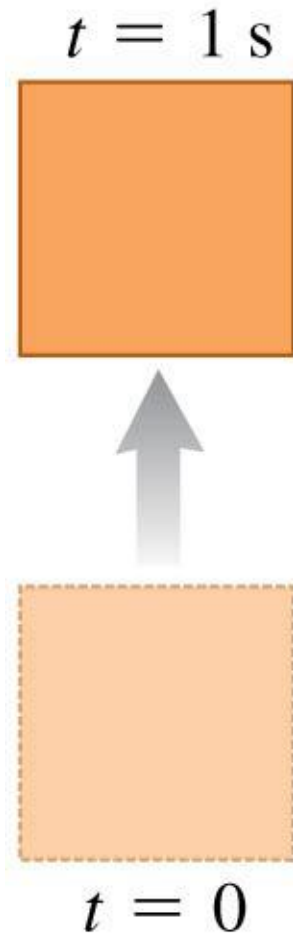
Work you do on the box
to lift it in 5 s:

$$W = 100 \text{ J}$$

Your power output: (average)

$$P = \frac{W}{t} = \frac{100 \text{ J}}{5 \text{ s}} = 20 \text{ W}$$

Power: Lifting a box quickly



Work you do on the same box to lift it the same distance in 1 s:

$$W = 100 \text{ J}$$

Your power output: (average)

$$P = \frac{W}{t} = \frac{100 \text{ J}}{1 \text{ s}} = 100 \text{ W}$$

Power

- In mechanics we can also express *instantaneous* power in terms of force and velocity:

Instantaneous power
for a force doing work
on a particle

$$P = \vec{F} \cdot \vec{v}$$

Force that acts on particle
Velocity of particle

Power: general formalism

Power is defined as the amount of **work** performed **per unit of time**.

If a machine or engine performs a certain amount of work, dW , within a given time interval, dt , the instantaneous power generated can be calculated as

$$P \equiv \frac{dW}{dt}$$

Since the work can be expressed as , the power can be written

$$P \equiv \frac{dW}{dt} = \frac{\vec{F} \cdot d\vec{r}}{dt} = \vec{F} \cdot \frac{d\vec{r}}{dt} = \vec{F} \cdot \vec{v}$$

Thus, power is a **scalar** defined as the product of the **force** and **velocity** components acting in the **same direction**.

Power

Using scalar notation, power can be written

$$P = \vec{F} \cdot \vec{v} = Fv \cos \theta$$

where θ is the angle between the force and velocity vectors.

So if the velocity of a body acted on by a force is known, the power can be determined by calculating the dot product or by multiplying force and velocity components.

Efficiency

The **mechanical efficiency** of a machine is the ratio of the **useful** power produced (**output power**) to the power supplied to the machine (**input power**) or

$$= (\text{power output}) / (\text{power input})$$

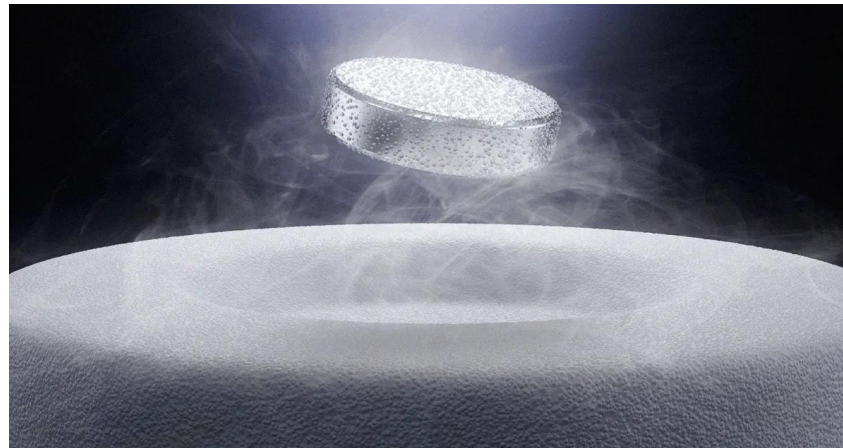
If energy input and removal occur at the same time, **mechanical efficiency** may also be expressed in terms of the ratio of **output energy** to **input energy** or

$$= (\text{energy output}) / (\text{energy input})$$

Efficiency

Machines will always have frictional forces. Since frictional forces **dissipate** energy, additional power will be required to overcome these forces. Consequently, the efficiency of a machine is always **less than 1**.

Superconductor



Procedure for analysis

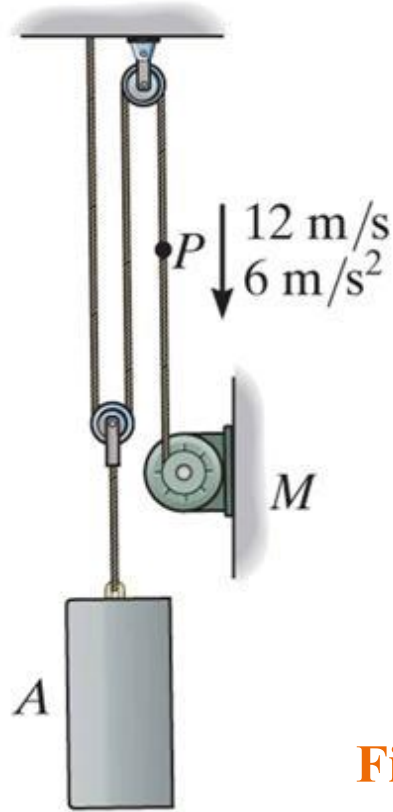
- Find the **resultant external force** acting on the body causing its motion. It may be necessary to draw a free-body diagram.
- Determine the **velocity** of the **point** on the body **at which the force is applied**. Energy methods or the equation of motion and appropriate kinematic relations, may be necessary.
- Multiply the **force magnitude** by the component of **velocity** acting **in the direction** of $\mathbf{F}$ to determine the power supplied to the body ($P = F v \cos \theta$).
- In some cases, **power** may be found by calculating the **work done per unit of time** ($P = dW/dt$).
- If the **mechanical efficiency** of a machine is known, either the power input or output can be determined.

Quiz

1. The formula definition of power is _____.
A) dW / dt B) $\mathbf{F} \cdot \mathbf{v}$
C) $\mathbf{F} \cdot d\mathbf{x}/dt$ D) All of the above.

2. Kinetic energy results from _____.
A) displacement B) velocity
C) gravity D) friction

Example (Atwood machine)



Given: A 50-kg block (*A*) is hoisted by the pulley system and motor *M*. The motor has an efficiency of 0.8. At this instant, point *P* on the cable has a velocity of 12 m/s which is increasing at a rate of 6 m/s^2 . Neglect the mass of the pulleys and cable.

Find: The power supplied to the motor at this instant.

Quiz

A twin engine jet aircraft is climbing at a 10 degree angle at 78 m/s. The thrust developed by a jet engine is 5000 N. The power developed by the aircraft is

A) $(5000 \text{ N})(78 \text{ m/s})$

B) $(10000 \text{ N})(78 \text{ m/s}) \cos 10^\circ$

C) $(5000 \text{ N})(78 \text{ m/s}) \cos 10^\circ$

D) $(10000 \text{ N})(78 \text{ m/s})$

Summary

- Work
- Kinetic energy
- The work-energy theorem
- Power
- Mechanical efficiency